Can you guess why did the covered candle extinguish?

_____

Let us summarize the happenings.

The candle 1 continues to burn, unless it is blown – off by strong moving air or any other external force. This is because fresh air is continuously available to the candle for its burning process.

Candle 2 glows for a while and then gets put – off. When the burning candle is covered with a glass tumbler, the candle can use the oxygen available in the air inside the glass tumbler. Since only a small amount of air is present inside the glass tumbler – only a small portion of oxygen is available for the candle to continue glowing. When all the oxygen of the air inside the gas jar is used up, then the burning candle gets extinguished.

Now, repeat the candle – glowing experiment taking four containers of different sizes. For example, you can take a 250ml conical flask, a 500ml bottle, a one – litre jar, a two – litre jar. Cover the burning candle one by one with these containers and find out how long it takes for the candle to extinguish in each case. Record your observations in the following table.

S. No.	Volume of the container (ml)	Time taken for candle to extinguish (second)

Can you write interpretation based on your observations at the table?

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4.5 Importance of air for survival of plants and animals

Respiration in plants

Plants require energy for their growth and hence respiration also occurs in plants. During respiration, plants take in oxygen and release carbon–di–oxide, just as animals do. Gaseous exchange with air in atmosphere takes place in plants with the help of tiny holes called stomata present on their leaves.

Photosynthesis

Plants manufacture food by a process called photosynthesis. During photosynthesis, carbon-di-oxide from the air and water from the soil react in the presence of sunlight to produce food. Most plants possess a green pigment called chlorophyll and it is also used-up in the process of photosynthesis. The word equation given below explains the process of photosynthesis.

$$\text{Carbon-di-oxide + water} \xrightarrow[\text{Chlorophyll}]{\text{Sunlight}} \text{Food + Oxygen}$$

Plants release oxygen during photosynthesis which is much more than the oxygen consumed by the plants, during respiration.

Respiration in Animals

When we breathe in air, the oxygen present in the air reacts chemically with digested food within the body to produce carbon-di-oxide gas, water vapour and energy.

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